

# Urša Uršič, PhD (she/her)

## Postdoctoral researcher in experimental cellular biophysics

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My scientific mission is to expand human knowledge about the basic principles of life using approaches at the interphase between physics and biology. I love to collaborate with people from various scientific backgrounds on novel projects and creatively solve problems along the way. My doctoral project includes understanding how material properties of the cytoplasm affect cell division mechanisms in large embryonic cells.

## Experience

📅 December 2024

### Postcourse research

**Stanford University**

📍 Stanford, CA, USA

Continuation of the MBL Physiology course project on biomineralization of ossicles in sea cucumbers. Investigated how curved surfaces affect microtubule self-organization and growth *in vitro* and in extracted cytoplasm.

Mentor: Manu Prakash

📅 Jun – August 2024

### Physiology course

**Marine Biological Laboratory**

📍 Woods Hole, MA, USA

Rotations: 1. Extremotolerancy of *A. Pullulans* (mentor: Amy Gladfelter), 2. Phagocytosis with surface modifications of macrophages and targets. (mentor: Daniel Fletcher) 3. Biomineralization of ossicles in sea cucumbers. (mentor: Manu Prakash)

📅 Jun – August 2022

### Internship

**Flatiron Institute, Simons Foundation**

📍 New York, New York, USA

Conducted computational research in microtubule self-organization using the innovative simulation tool, aLENS, developed within the group.

Mentors: Adam Lamson & Michael Shelley

📅 Nov 2021

### Predoc course

**Max Planck Institute of Molecular Cell Biology and Genetics**

📍 Dresden, Germany

A 4-week group research project on the shininess of Zebrafish. Conducted experimental as well as computational research on guanine crystal formation in iridophores in Zebrafish skin.

Mentors: Rita Mateus, Eric Geertsma, Carl Modes

📅 Jun – Jul 2019

### Internship

**Protein, Structure and Function Group, University of Queensland**

📍 Brisbane, Australia

Characterization of innate immune system protein aggregation (TLR4 and MAL) using fluorescence confocal spectroscopy (FCS) techniques and cell-free protein expression techniques. Improved and automated analysis of the FCS data output.

Mentor: Boštjan Kobe

📅 Oct 2018 – Apr 2019

### Internship

**ESMP – University of Luxembourg**

📍 Luxembourg, Luxembourg

Laser-induced motion of topological defects on nematic liquid crystal shells. Production of liquid crystal shells with microfluidics and using a custom-made double confocal microscope imaging.

Mentor: Jan Lagerwall

📅 Feb 2017 – Sep 2018

### Student project

**K5, F5 – Institute 'Jozef Stefan'**

📍 Ljubljana, Slovenia

Characterization of the hydrodynamics of drying droplets for inkjet printing of flexible electronic circuits.

Production of metal oxide thin films and characterization of their temperature-dependent dielectric constant.

Mentor: Aleksander Matavž

## Education

📅 Oct 2021 – Jun 2026

**Ph.D. in Biophysics (summa cum laude)**

**Max Planck Institute of Molecular Cell Biology and Genetics, TU Dresden**

📍 Dresden, Germany

📅 Oct 2019 – Sep 2021

**M.Sc. in Biophysics**

**University of Ljubljana**

📍 Ljubljana, Slovenia  
grade 9.5/10

📅 Oct 2015 – Apr 2019

**B.Sc. in Physics**

**University of Ljubljana**

📍 Ljubljana, Slovenia  
grade 8.6/10

## Teaching & Mentoring

📅 Winter semester 2024

**Advanced Practical: Biomolecular motors**

**3rd year Physics students**

📍 TU Dresden, B-Cube

📅 June 2023

**Biophysical modelling workshop**

📍 Flatiron institute, New York  
Teaching assistant for aLENS (1/3 of the full course)

📅 Winter semester 2022

**Advanced Practical: Biomolecular motors**

**3rd year Physics students**

📍 TU Dresden, B-Cube

📅 Winter semester 2022

**Molecular Machines Practical**

**Masters course in Bioengineering**

📍 TU Dresden, MPI-CBG

📅 2016-2018

**Private saxophone lessons**

📍 Slovenija

## Publications

- Kickuth A., Uršič, U., Staddon, M., Brugues J., 2026, *Nature*
- Matavž, A., Uršič, U., Močivnik, J., Richter, D., Humar, M., Čopar, S., Malič, B. and Bobnar, V., 2022. *Journal of Colloid and Interface Science*, 608, pp.1718-1727.

## Talks & Presentations

- 2026 EMBO Workshop @Barcelona talk
- 2024 ASCB poster
- 2024 ASCB talk
- 2024 POL seminar
- 2023 Internal MPI-CBG seminar
- 2023 Invited talk at Slovene Biophysical Society
- 2023 Chalk talk at Science on Board at MPI-CBG
- 2022 Flatiron Institute summer internship poster

## Awards & Fellowships

- ★ 2024 Marine Biological laboratory postcourse research fellowship
- ★ 2024 Society for General Physiology Award for Scholars at MBL
- ★ 2024 CoolLED image competition
- ★ 2022 Simons Foundation summer intern fellowship
- ★ 2021 Dean's Award for top 5% of students at the Faculty of Mathematics and Physics, University of Ljubljana
- ★ 2011-2021 National 'Zois' scholarship for gifted students
- ★ 2021 Silver national award in competition in logics
- ★ 2020 BioCamp 2020 Fellow (Lek, Novartis)
- ★ 2019 Silver National Award in logics competition
- ★ 2019 ASEF Junior Fellowship

## Side projects

- Member of VTIS (2023-recent),
- Volunteer at Science goes to school (2023-2024)
- Volunteer at Moja Mavrica - safe space for the LGBT community with a focus on mental health (2022-2023),
- Volunteer at DIH - Peer support for LGBT youth (2021-2023),
- An active member of ASEF - buddy program, tutoring group, activities (2019-2022),
- Clean water round table, ASEF (2021)

## Hobbies

Music (saxophone, piano)

Dancing

Hiking

Running

Scuba diving

Drag performance

Volleyball